

A circle of radius 6 centered at (6, 0) is drawn.

Points are located at four places on the circle and labeled O, P, Q, and R in clockwise order.

- O = (0, 0)
- R = (12, 0)
- Angle POQ = Angle QOR = x
- Chord OQ = 9 units
- Find: Length of chord OP

Step 1: Use triangle OQR to find $\cos(x)$ and $\sin(x)$

- Triangle OQR is a right triangle with hypotenuse OR = 12
- OQ = 9

Using the Pythagorean Theorem:

$$QR = \sqrt{12^2 - 9^2} = \sqrt{144 - 81} = \sqrt{63}$$

So:

- $\cos(x) = 9 / 12 = 3/4$
- $\sin(x) = \sqrt{63} / 12$

Step 2: Use the double angle identity

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$= (3/4)^2 - (\sqrt{63}/12)^2$$

$$= 9/16 - 63/144$$

Convert to common denominator:

$$9/16 = 81/144$$

Then:

$$\cos(2x) = 81/144 - 63/144 = 18/144 = 1/8$$

Step 3: Use the relation

$$\cos(2x) = OP / 12$$

$$1/8 = OP / 12$$

$$OP = 12 * (1/8) = 1.5 \text{ units}$$

? Final Answer:

Chord OP is **1.5 units long**